JAVIER LOPEZ

Engineering Design Portfolio

A curated collection of mechanical engineering, product design, AI-driven workflows, and prototyping projects.

Cover Page

JAVIER LOPEZ
Mechanical Engineering (M.S)- Design & Manufacturing
San José State University (MSME, 2026)
(510) 940-3119
javier.lopezh15@gmail.com
LinkedIn: https://www.linkedin.com/in/francisco1-javier2-lopez0/

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1. ABOUT ME

Mechanical Engineer (M.S. SJSU) with hands-on experience spanning aerodynamic design, carbon fiber composite fabrication, robotics, and AI-driven engineering workflows. I've delivered production hardware at Lockheed Martin, built a first-of-its-kind multi-modal quadruped robot for NASA lunar exploration, and developed full ML-to-CFD pipelines that bridge generative AI with physics-based simulation. My work lives at the intersection of mechanical design, advanced manufacturing, and intelligent automation, I don't just model parts, I build and validate them. Currently competing with SJSU Formula SAE as an aerodynamics engineer while pursuing graduate research in AI-assisted vehicle design.

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2. TECHNICAL EXPERTISE

CAD & Simulation: SolidWorks, PTC Creo Parametric, Fusion 360, CATIA, Onshape, ANSYS Fluent (CFD/FEA)

Programming & Autonomy: Python, MATLAB, C++, C, ROS, autonomous trajectory algorithms

AI & Computer Vision: OpenCV, TensorFlow/Keras, scikit-learn, CNN development, ML classification, generative AI, prompt engineering

Robotics & Mechatronics: Arduino, Raspberry Pi, brushless DC motor control, planetary gearing, LiDAR & ultrasonic sensors, sensor integration

Manufacturing: CNC machining, 3D printing (FDM), carbon fiber composite layup, vacuum bagging, GD&T, DFM/DFA, post-cure finishing, welded aluminum fabrication, AR500 steel fabrication

Aerodynamics: ANSYS Fluent (k-ω SST), external flow simulation, drag coefficient analysis, pressure/velocity contour post-processing, parametric geometry optimization

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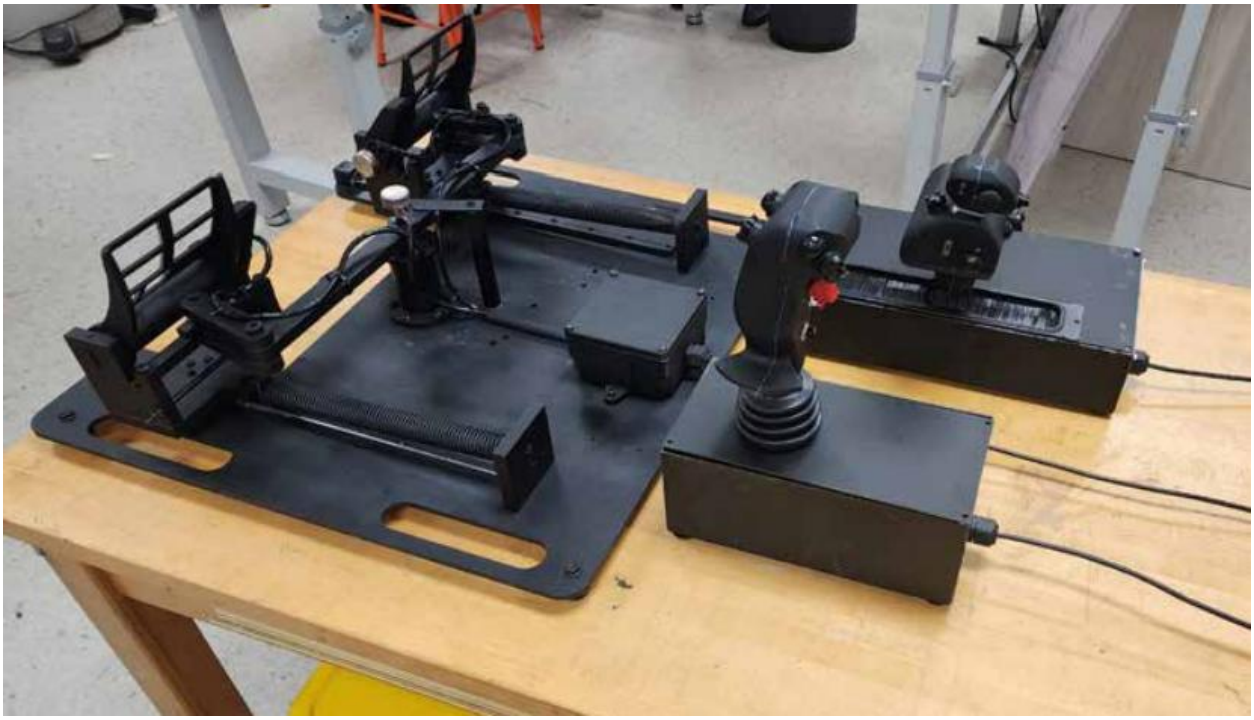
3. PROJECT PORTFOLIO

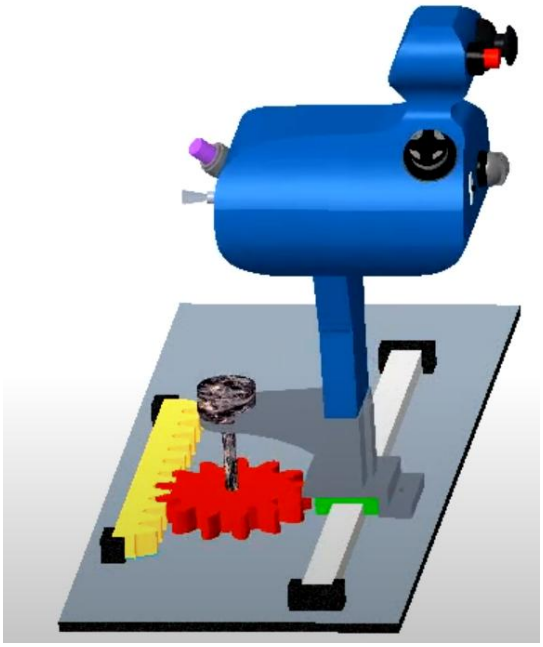
3.1 Mechanical Design Projects

F-35 Simulated Throttle & Joystick - Lockheed Martin

Description: Designed and delivered a low-cost F-35 flight simulator control system for Lockheed Martin, comprising a fully functional joystick, throttle, and rudder pedal system built to Air Force ergonomic standards (5'3"–6'4" operator range) within a $1,272 budget. The joystick featured a multi-plane gimbal mechanism with a carbon steel wave spring for realistic stick resistance, while the throttle used a rack-and-pinion linear rail for smooth, repeatable actuation. Structural integrity was validated through FEA in PTC Creo Parametric under operational loads of 7.5 lbf (joystick) and 15 lbf (throttle) before any hardware was produced. All components were 3D printed, post-processed, painted, and assembled in-house, with a custom PCB shared across all three subsystems to simplify manufacturing and reduce cost. Authored full GD&T documentation with exploded assembly views to enable clean manufacturing handoff and future maintenance.

Skills Used: PTC Creo Parametric, FEA, GD&T documentation, gimbal mechanism design, rack-and-pinion actuation, 3D printing (FDM), PCB integration, post-processing & painting, ergonomic design, budget management, assembly documentation.





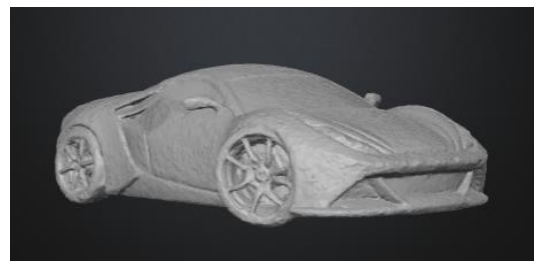
3.2 Graduate Research

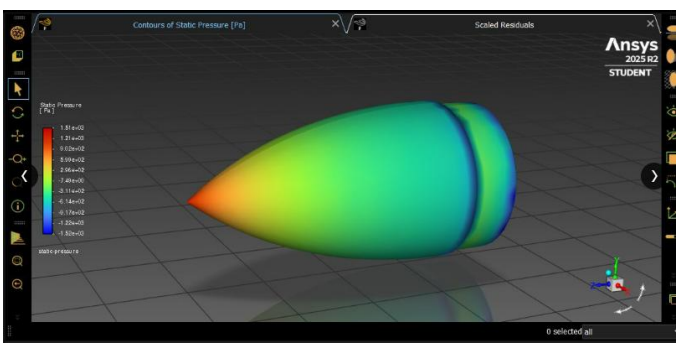
From Generative Geometry to Simulation-Ready Models: An AI-Assisted Design-to-CFD Workflow for Aerodynamic Analysis

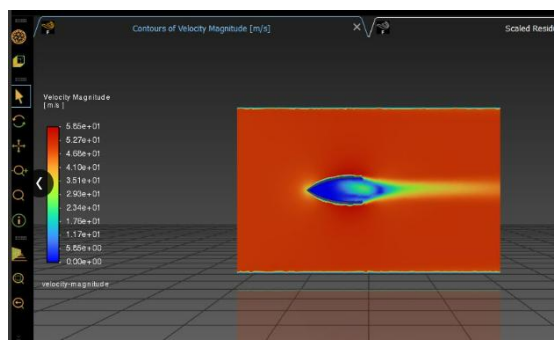
Description: This research develops an AI-assisted design-to-CFD workflow that integrates generative geometry, automated geometry repair, and high-fidelity aerodynamic simulation. The goal is to address a key challenge in engineering workflows: converting unconstrained AI-generated shapes into simulation-ready models that can be validated using Computational Fluid Dynamics (CFD) and Finite Element Analysis (FEA).

The pipeline combines generative CAD tools with geometry validation and ANSYS Fluent simulations to evaluate aerodynamic performance. The workflow was tested on external flow problems including rocket nose cone and vehicle-inspired geometries, analyzing pressure distribution, velocity fields, and drag coefficient predictions. Results produced physically consistent aerodynamic behavior and demonstrated how AI can accelerate early-stage design exploration while maintaining physics-based validation.

Skills Used: CAD modeling, generative geometry design, Python, Computational Fluid Dynamics (CFD), ANSYS Fluent, turbulence modeling (k-ω SST), geometry repair and mesh preparation, aerodynamic analysis, drag coefficient evaluation, simulation validation, parametric design, design optimization, and engineering data analysis.







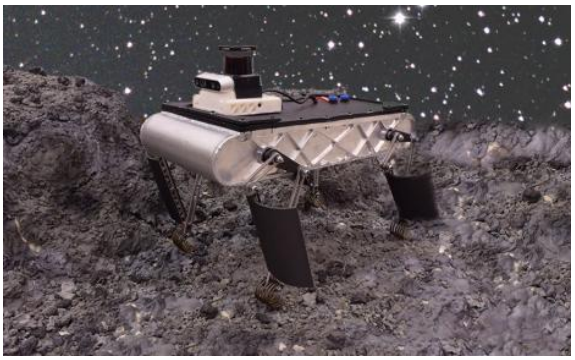
3.3 Robotics & Mechatronics Projects

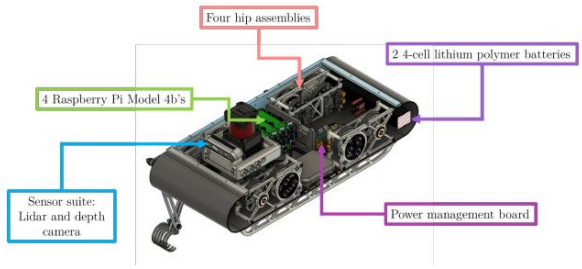
Extreme Terrain Quadruped (ET-Quad)

Description: Designed and built a quadruped robot for NASA lunar exploration capable of walking (25 cm/s), 90° vertical climbing, and swimming from a single hardware configuration,

the first legged robot to achieve all three modes without reconfiguration. The mechanical design centered on coaxial hip assemblies with 6:1 planetary gearing and eight brushless DC motors coordinated via distributed ROS control. A custom AR500 steel foot solved the competing demands of vertical surface attachment and walking compliance in one part. Underwater operation required an ISO 22810-compliant waterproof enclosure with shaft seals and a welded aluminum chassis. A Python-based trajectory algorithm automated mode transitions, cutting manual intervention by 30%. Performance validated on custom boom arms, a drag tank, and climbing testbeds.

Skills Used: SolidWorks, Python, ROS, brushless DC motor control, sensor integration, AR500 steel fabrication, autonomous trajectory algorithms, performance testbed validation.





3.4 AI-Driven Engineering & Simulation Projects

CNN Classifier for Car Front-End Designs

Description: Developed a Convolutional Neural Network (CNN) to classify vehicle types from images, including categories such as sports cars, trucks, and minivans. The dataset consisted of labeled vehicle images that were preprocessed using Python, including resizing, normalization, and splitting into training, validation, and test sets. The CNN architecture used convolutional and max-pooling layers to extract visual features and fully connected layers for classification. Model performance was evaluated using accuracy, loss curves, and confusion matrix analysis, demonstrating reliable classification across vehicle categories.

Skills/ Tools: Python, TensorFlow, Keras, Convolutional Neural Networks (CNNs), image classification, dataset preprocessing, training/validation/test dataset splitting, deep learning model training, accuracy and loss evaluation, confusion matrix analysis, and machine learning model performance evaluation.

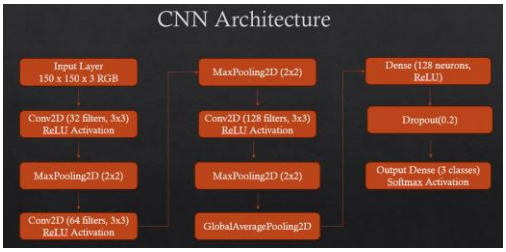


Table 1: Quantitative Evaluation Metrics for AI-Generated Vehicle Front-End Designs Obtained from Python Image Analysis Script

Vehicle Type	Bounding Box Area (pixels²)	Silhouette Area (pixels²)	Efficiency Ratio	Aspect Ratio	Symmetry Score	Processing Time (s)	Symmetry Time (s)
Sports Car	28060	22395	0.8	3.32	0.96	5.9849	3.6078
Truck	50736	25800	0.51	1.8	0.79	2.8981	1.3098
Minivan	50820	32855	0.65	1.87	0.96	1.5181	2.1115

Bounding Box Visualization

Symmetry Difference (L vs Flipped R)

3.5 Formula SAE Projects

Carbon Fiber Composite Manufacturing for Formula SAE Car

Description: As part of Spartan Racing (SJSU Formula SAE), I executed full end-to-end carbon fiber composite fabrication for aerodynamic components including front wing elements, undertray panels, and mold tooling. The process began with pattern cutting of 2×2 twill and plain-weave carbon fiber fabric to net shape, maintaining precise fiber direction alignment per design intent. I performed multi-ply wet hand layups with controlled epoxy resin application, consolidating plies with squeegees to achieve uniform resin-to-fiber ratio and eliminate dry spots. Each part was vacuum bagged with full peel ply and breather cloth assemblies, with thermocouple-monitored seal integrity to maintain consistent vacuum throughout cure. Components were then staged in an industrial oven with active vacuum pull maintained during the full cure cycle per resin manufacturer specifications. Post-cure operations included trimming, edge cleanup, and surface sanding to meet vehicle integration tolerances and aerodynamic surface requirements. This work built hands-on proficiency in composite ply scheduling, vacuum bag assembly and leak detection, cure cycle management, mold surface preparation, and dimensional verification for fit and finish on a competition vehicle.

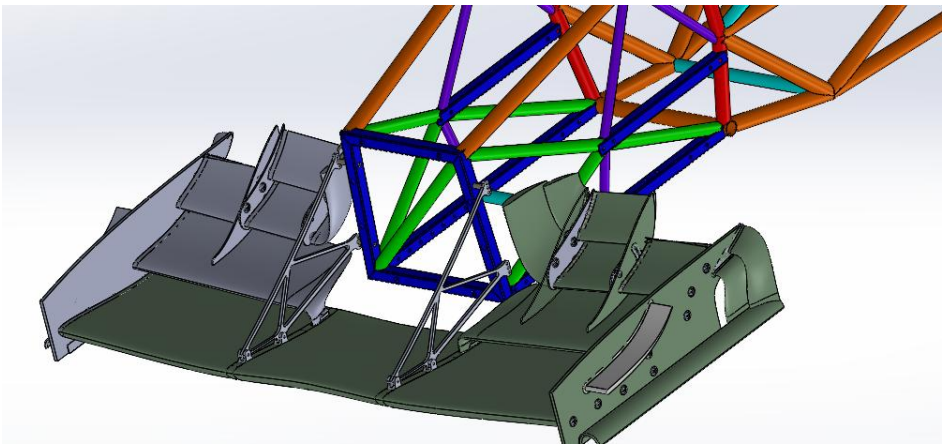
Skills: Wet layup, vacuum bagging, peel ply & breather cloth assembly, thermocouple-monitored seal integrity, oven cure cycle management, composite ply scheduling, epoxy resin application, mold surface preparation, post-cure trimming & sanding, dimensional verification, carbon fiber fabric cutting.

Aerodynamic Component Design & Manufacturing Jigs

As part of the aerodynamics team at Spartan Racing, I contributed to the design and validation of front wing brackets and manufacturing jigs for the FSAE competition vehicle. The process began by reviewing past team documentation and SOPs to understand existing bracket geometry and design intent, then performing FEA on updated bracket designs to validate structural integrity under race loads before committing to fabrication. Once new bracket dimensions were finalized,

I designed precision alignment jigs in SolidWorks to fix the front wing brackets during assembly, ensuring correct positioning and repeatability across the car. GD&T was applied throughout to define tight tolerances and ensure the jigs held components securely without movement under dynamic vehicle loads. The goal was eliminating assembly variability, if a jig fits correctly, the bracket is in the right place every time.

Skills: SolidWorks, FEA, GD&T, jig & fixture design, bracket design, engineering documentation review, SOP compliance, tolerance analysis, assembly validation.



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4. ADDITIONAL SKILLS & ACHIEVEMENTS

- Engineering leadership & team collaboration
- AI-augmented design and automation
- Technical documentation & presentation skills
- Hands-on fabrication and prototyping
- Full ML-to-CAE pipeline development (generative AI → CFD validation)
- Multi-domain engineering: mechanical, software, aerodynamics, and composites
- Budget management & hardware delivery: $1,272 production budget at Lockheed Martin
- Cross-disciplinary research: M.S. candidate bridging AI, CFD, and mechanical design
- Competition engineering: Formula SAE design, fabrication, and vehicle integration
- Bilingual: English & Spanish